

i Examination in IN3120/IN4120

UNIVERSITY OF OSLO

The Faculty of Mathematics and Natural Sciences

Written examination IN3120/IN4120

2025 Autumn

Duration: December 1, 15:00 - 19.00 (4 hours)

Permitted aids: Inspira calculator available

It is important that you read this front page before you start.

The different questions have different weights, as indicated.

You can answer in Norwegian or English. Please use the language that you are most comfortable with.

1 NEAREST NEIGHBORS I [10p]

Below are five statements about the k -nearest neighbor (k -NN) method. Classify each statement as being either true or false and provide a clear justification for your answer. Answers without justifications will give no points. Unless otherwise noted, you can assume a standard (unweighted) k -NN with Euclidean distance.

(a) [2p] “The decision boundaries are piecewise linear.”

(b) [2p] “Increasing k always reduces the training error.”

(c) [2p] “Standardizing or normalizing features can change the resulting classification.”

(d) [2p] “Choosing the value of k is typically done using the test set to maximize accuracy.”

(e) [2p] “When two or more classes are equally represented among the k nearest neighbors, the classifier’s prediction is always undefined unless a tie-breaking rule is specified.”

Write your answer in the box below. Changes are saved automatically.

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Words: 0

Maximum marks: 10

2 NEAREST NEIGHBORS II [15p]

Below are four different approaches for implementing an approximate nearest neighbor (ANN) index. For each approach, provide a short explanation of its basic idea. Your explanations should touch on both how data is organized in the index, and how data is queried.

(a) [4p] Tree-based algorithms

(b) [4p] Locality-sensitive hashing

(c) [3p] Quantization or clustering-based algorithms

(d) [4p] Graph-based algorithms

Write your answer in the box below. Changes are saved automatically.

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Words: 0

Maximum marks: 15

(c) [8p] Postings in a postings list can store additional information along with document identifiers. List three types of additional information that can be stored and discuss how each can improve query processing and/or relevance. Listing an item without discussion will give no points.

Write your answer in the box below. Changes are saved automatically.

Maximum marks: 20

